

Yash Agarwal

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EDUCATION

Carnegie Mellon University, School of Computer Science

Pittsburgh, PA

Masters in Intelligent Information Systems - GPA - 3.9/4.0

December 2026

- Coursework - Computer Systems, DL Systems, LLM Systems, Deep Learning, Advanced NLP, Multimodal ML
- Advisor - Prof. **Tianqi Chen** (NVIDIA, creator of TVM, XGBoost) - [Machine Learning Compiler](#)

Indian Institute of Technology (IIT), Kharagpur

Kharagpur, India

Bachelors in Computer Science and Engineering - GPA 8.65/10

May 2023

- **All India Rank 250** out of 1.2 million applicants for Joint Entrance Examination (JEE Advanced)
- Relevant Coursework - Deep Learning, Computer Architecture, Operating Systems, Compilers, Computer Networks
- Publication - YesBut: Evaluating Vision-Language Models

EMNLP 2024

WORK EXPERIENCE

Modular Inc

San Francisco, CA

AI Performance Engineering Intern

May 2026 - Present

- Accelerated pinned (page-locked, DMA-capable) host-memory allocation by **10x** for tiered **KV-cache offload**, cutting a 1.5TB allocation from 50+ minutes to under 5, unblocking **trillion-parameter MoE** serving on 8-GPU nodes
- Built a **lock-free MPSC** pipeline that faults pages across NUMA-pinned threads at 99.2 GB/s, isolates per-region fault locks via low-level Linux VM **syscalls** to eliminate lock contention, and **pipelines** it with chunked GPU registration

Squarepoint Capital

Bangalore, India

Software Developer C++

July 2023 - July 2025

- Architected new trading capabilities for FX Options, implementing end-to-end solution in **C++ 20** and **Python** using **STL**, **object oriented design** and comprehensive **tests**, that pioneered new asset class capability at the firm
- Redesigned messaging protocol by refactoring 2000+ line C++ protocol into modular object oriented structure
- Debugged critical production crashes using **gdb** core dump analysis and implemented **thread-safety** in memory access
- Redesigned **multithreaded** Python testing framework to single-threaded architecture eliminating **40%** test flakes

PROJECTS @ CARNEGIE MELLON

Malloc, cache, shell | Computer Systems Course Projects

Spring 2026

- Implemented **malloc** using **mmap** and free-lists - implicit, explicit, segregated, block coalescing and various fit policies
- Optimized **throughput** using Better Fit policy and memory **utilization** using **seg-lists** with miniblocks and no footers
- Implemented a **cache** simulator, optimized matrix transpose by tiling and eliminating **evictions**, 4th in a class of 200
- Implemented unix shell with signal handling, bg/fg commands, forking and reaping child processes and IO redirection
- Parallelized a File System with **RW locks**, enabling **concurrent** reads and writes to shared inodes, blocks, directories

Needle - Optimizing CUDA GPU Kernels and Autodiff | DL Systems Course Project

Fall 2025

- Optimized Matmul **CUDA kernel** by tiling, vectorization and GMEM coalescing, achieving 85% performance of **cuBLAS**
- Implemented **Flash Attention**, including kernel fusion, tiling, online softmax, achieving 77% performance of PyTorch
- Architected automatic differentiation Tensor, complete with mathematical ops and nn.Modules for Conv, RNN, MHA
- Developed NDArray class supporting reshaping, broadcasting, summation, Matmul using **C++** and **CUDA** backend

WhisQer - Quantization trained Conformer | Deep Learning Course Project

Fall 2025

- Architected **quantized** Conformer ASR with 2-bit and 1-bit precision variants, with learnable scaling parameters, compressing a 108MB FP32 model to 45.6MB 2-bit by quantizing encoder FFN and attention projections
- Implemented teacher student co-training combining CTC and KL divergence loss and straight through estimator
- Built end-to-end PyTorch ASR training pipeline with streaming audio dataloader, feature pipeline, and CTC decoding

SKILLS

Programming Languages and Tools - C++ 23, Python, Rust, Bash, Git, Gitlab, Linux, Bazel, CMake, STL

Machine Learning - PyTorch, JAX, Numpy, CUDA, Triton, Mojo, Wandb, HuggingFace